EXPERIMENT 5

(i) Analysis of Fourier Series Representation of % Continuous-Time Periodic Signals

```
clc;

clear;

 close all;

figure;

% Fourier Series Representation of a Periodic Signal A = 1; f0 = 1; T = 1/f0;

t = linspace(0, 2*T, 2000);

% Half-Wave Rectified Sinusoidal Signal x = Asin(2pif0t); x(x < 0) = 0;

% Number of Harmonics N = 10;

% Fourier Series Coefficients a0 = (1/T)*trapz(t, x)/2; an = zeros(1, N); bn = zeros(1, N);

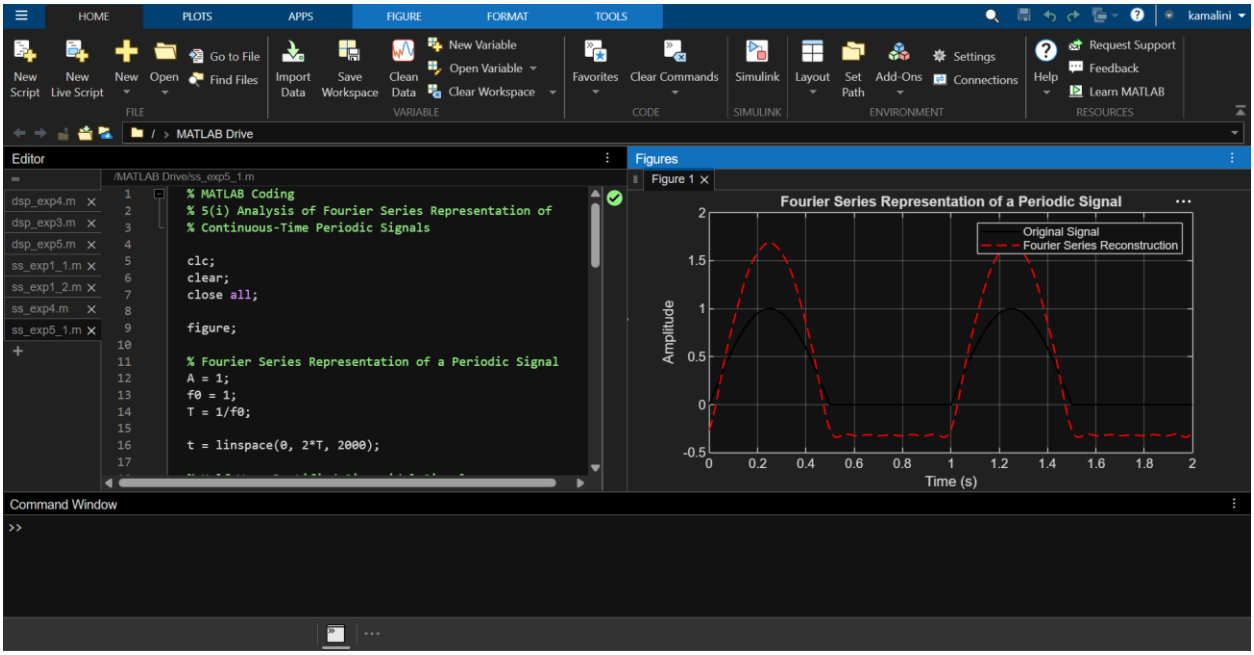
for k = 1:N an(k) = (2/T)trapz(t, x.cos(2pikf0t)); bn(k) = (2/T)trapz(t, x.sin(2pikf0t)); end

% Fourier Series Reconstruction xFS = a0*ones(size(t));

for k = 1:N xFS = xFS + an(k)cos(2pikf0t) ... + bn(k)sin(2pikf0t); end

% Plotting plot(t, x, 'k', 'LineWidth', 1.2); hold on; plot(t, xFS, 'r--', 'LineWidth', 1.2);

grid on; xlabel('Time (s)'); ylabel('Amplitude'); title('Fourier Series Representation of a Periodic
Signal'); legend('Original Signal', 'Fourier Series Reconstruction');
```



(iv) Verification of Fourier Series Reconstruction and Analysis of Signal Characteristics

clc; clear; close all;

figure;

% Fourier Series Reconstruction and Error Analysis f0 = 1; T = 1/f0; t = linspace(0, 2*T, 4000);

% Half-Wave Rectified Sinusoidal Signal x = max(sin(2*pi*f0*t), 0);

% Number of Harmonics N = 10;

% Extract One Period tt = t(1:2000); xx = x(1:2000);

% Fourier Series Coefficients a0 = (2/T)*trapz(tt, xx);

an = zeros(1, N); bn = zeros(1, N);

for k = 1:N an(k) = (2/T)*trapz(tt, xx.*cos(2*pi*k*f0*tt)); bn(k) = (2/T)*trapz(tt, xx.*sin(2*pi*k*f0*tt)); end

% Fourier Series Reconstruction xr = a0/2*ones(size(t));

for k = 1:N xr = xr + an(k)*cos(2*pi*k*f0*t) ... + bn(k)*sin(2*pi*k*f0*t); end

% Reconstruction Error err = x - xr;

```
% Plot Original and Reconstructed Signals subplot(2,1,1);
plot(t, x, 'k', t, xr, 'r--', 'LineWidth', 1.1);
 grid on; xlabel('Time (s)'); ylabel('Amplitude');
 title('Original and Reconstructed Signals');
 legend('Original', 'Reconstructed');
% Plot Reconstruction Error subplot(2,1,2);
plot(t, err, 'b');
grid on;
 xlabel('Time (s)'); ylabel('Error');
 title('Reconstruction Error');
```

